

Cash-flow audit and bill-date optimisation

Sun 16 Aug to Wed 14 Oct - 60 days projected

Opening balance	\$2,414.80 (as entered 2026-08-16)
Projection window	Sun 16 Aug - Wed 14 Oct
Lowest projected balance	-\$4,227.34 on Tue 6 Oct
Days below zero	31 of 60
Bill-date changes the data supports	0
Obligations modelled	39 active bills, 2 income streams
Generated	17 Aug 2026 00:15

1. Executive summary

Balance now

\$2,414.80

Lowest projected

-\$4,227.34

Days negative

31/60

Moves available

0

Moving bill dates cannot fix this, and the report will not pretend it can

The projection goes below zero on 31 of 60 days and reaches -\$4,227.34. No payment-date change is recommended, for one reason: of the 17 obligations classified FIXED and the 24 classified UNKNOWN, none has a recorded latest-safe payment date. Section 4 lists exactly which ones to confirm and section 5 shows what confirming them would buy. Section 6 shows why the largest gap survives even then.

Across the 60 days projected, \$18,144.54 of income is set against \$22,259.90 of obligations. That is a net -\$4,115.36, or about -\$2,057.68 per 30 days. This is the number that decides everything else in this report: where income exceeds obligations, a cash-flow problem is a timing problem and moving payment dates can solve it. Where it does not, no schedule solves it, and the honest output is the size of the gap.

This is a shortfall, not a timing problem

Obligations exceed income by \$4,115.36 over the window (-\$2,057.68 per 30 days). Rescheduling moves a deficit around the calendar; it cannot remove one. Every bill could be paid on its most convenient possible date and the account would still end the window \$4,115.36 further behind. Sections 6 and 7 deal with this directly.

The risk dates, in order

Period	Days	Worst balance	On	Largest obligations in the period
24 Aug - 25 Aug	2	-\$303.09	24 Aug	Mortgage \$1,801.62, Kia \$742.65, Upstart past bills \$86.46
7 Sep - 8 Sep	2	-\$699.64	8 Sep	Mortgage \$1,801.62, PNC \$150.00
15 Sep - 15 Sep	1	-\$413.47	15 Sep	Loew & Patel braces \$400.00, Upstart debt consol \$161.00
19 Sep - 14 Oct	26	-\$4,227.34	6 Oct	Mortgage \$1,801.62, Mortgage \$1,801.62, Town of Clinton water \$873.53

What to do first

- Honour 5 household-corrected amount(s): would lift the lowest projected balance by \$1,005.50 to -\$3,221.84 and take negative days from 31 to 24. Confirm whether it is true.
- All variable bills at their stated amount: would lift the lowest projected balance by \$1,683.20 to -\$2,544.14 and take negative days from 31 to 21. Confirm whether it is true.
- Reactivate: Branchburg BOE payroll: would lift the lowest projected balance by \$5,275.09 to \$1,047.75 and take negative days from 31 to 0. Confirm whether it is true.
- Both corrections together: would lift the lowest projected balance by \$5,446.09 to \$1,218.75 and take negative days from 31 to 0. Confirm whether it is true.
- Supply a latest-safe payment date for the 24 obligations listed in section 4. Section 5 shows the schedule that becomes available once they are known.

| 2. Starting position and how it was derived

Every figure in this report runs forward from \$2,414.80, the most recent balance recorded in state.json. It was entered on 2026-08-16T19:49:34 and represents 2026-08-16. It is a single figure typed in by the household, not a bank feed: there is no connection to the bank in this system, so pending transactions, uncleared cheques and holds are included only to the extent the bank had already applied them when it was read.

Accounts, transfers and double counting

The model holds one pooled checking balance. There is no second account, no savings sweep and no internal transfer in bills.json or income.json, so no transfer can be miscounted as income or as an expense. That is a genuine simplification rather than a verified fact about the household: if money is regularly moved to or from another account, none of it is modelled here. config.json does exclude transfer-like descriptions (transfer, xfer, zelle, venmo, cash app and others) when classifying spending, which is the one place the distinction is currently drawn.

One double-count risk is real and worth stating. The balance was entered at 19:49 on 2026-08-16, near the end of that day, while the projection charges that day's bills in full. Anything that had already cleared before it was typed in is therefore subtracted twice. The engine has no way to detect this: it has no transaction feed, only a number and a date.

3. Income model

Income is modelled on actual arrival dates, never as a monthly average. A weekly deposit lands on its weekday and is available that day; a biweekly one steps 14 days from its anchor. This matters because the whole question is what is in the account on a given morning.

Source	Amount	Cadence	Anchor	Counted	Times	Total in wind...	Status
Branchburg BOE payroll	\$2,829.31	biweekly	2026-06-10	NO - switched off	0	\$0.00	unverified
Insight Global payroll	\$2,016.06	weekly	2026-07-08	yes	9	\$18,144.54	unverified
Charlie and Clark	\$510.00	once	2026-08-13	yes	0	\$0.00	unverified

Totals cover 16 Aug to 14 Oct. 'Times' is the number of deposits the calendar generates in that period.

1 income source(s) are recorded but not counted

Branchburg BOE payroll (\$2,829.31 biweekly) is switched off in income.json. Note on file: Edited in the app 13 Aug 2026: deactivated. Previous note: Amount corrected from \$2,840.34. Observed \$2,829.31 on 04/13, 04/28, 05/13, 05/27 and 06/10/2026, exactly 14-15 days apart. A \$2,837.55 credit on 06/16 breaks the pattern and may be a summer-schedule payment - verify before the school year s

4. Bill movability - which dates may legitimately change

A payment date can only be moved if something establishes how late is still safe. Three classifications are used, and the rule for each is stated so it can be checked rather than trusted.

- **FIXED** - secured against an asset, or recorded as non-deferrable. Never moved later by this report.
- **FLEXIBLE** - a `payment_window` is recorded in `bills.json` giving the earliest and latest safe dates. Only these can be optimised on evidence.
- **UNKNOWN - USER CONFIRMATION REQUIRED** - marked deferrable, but no due date, grace period or late-fee term is recorded, so how far it may slip is not derivable. Never moved later in section 5's baseline.

Paying **EARLIER** is treated as always permissible, because paying ahead of schedule is never late. It is bounded at one cadence so a bill cannot be pulled on top of its own previous occurrence. Moving earlier is what needs no confirmation; moving later is what does.

Bill	Forecast	Freq	Next due	Tier	Class	Earliest safe	Latest safe	Basis for the classification
CCU car loan	\$306.90	monthly	18 Aug	1	FIXED	22 Jul	not established	Secured against an asset. A late payment risks repossession or foreclosure, so this is never moved later.
Kia	\$742.65	monthly	24 Aug	1	FIXED	28 Jul	not established	Secured against an asset. A late payment risks repossession or foreclosure, so this is never moved later.
Mortgage	\$1,801.62	biweekly	24 Aug	1	FIXED	11 Aug	not established	Secured against an asset. A late payment risks repossession or foreclosure, so this is never moved later.
Comcast	\$138.93	monthly	18 Aug	2	UNKNOWN	22 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Elizabethtown Gas	\$60.01	monthly	11 Sep	2	FIXED	15 Aug	not established	Recorded as non-deferrable in <code>bills.json</code> - missing it has a consequence the household has already flagged.
Fuel	\$90.77	weekly	16 Aug	2	FIXED	10 Aug	not established	Recorded as non-deferrable in <code>bills.json</code> - missing it has a consequence the household has already flagged.
Groceries	\$319.09	weekly	16 Aug	2	FIXED	10 Aug	not established	Recorded as non-deferrable in <code>bills.json</code> - missing it has a consequence the household has already flagged.
Hanover Auto	\$513.27	monthly	12 Sep	2	FIXED	16 Aug	not established	Recorded as non-deferrable in <code>bills.json</code> - missing it has a consequence the household has already flagged.

Bill	Forecast	Freq	Next due	Tier	Class	Earliest safe	Latest safe	Basis for the classification
IRS federal	\$103.00	monthly	19 Aug	2	FIXED	23 Jul	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
IRS state	\$114.00	monthly	19 Aug	2	FIXED	23 Jul	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
JCP&L / FirstEnergy	\$374.22	monthly	3 Sep	2	FIXED	7 Aug	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
Sewer	\$304.50	annual	1 Sep	2	FIXED	2 Sep	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
T-Mobile Lia	\$20.75	monthly	11 Sep	2	UNKNOWN	15 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Town of Clinton water	\$873.53	quarterly	19 Sep	2	FIXED	22 Jun	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
Xfinity Mobile	\$88.82	monthly	29 Aug	2	UNKNOWN	2 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Affirm	\$73.07	monthly	29 Aug	3	UNKNOWN	2 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Barclay Old Navy	\$84.78	monthly	27 Aug	3	UNKNOWN	31 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Capital One CC	\$163.00	monthly	17 Aug	3	UNKNOWN	21 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Chase CC Amazon	\$136.00	monthly	17 Aug	3	UNKNOWN	21 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Chase CC United	\$88.00	monthly	22 Aug	3	UNKNOWN	26 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.

Bill	Forecast	Freq	Next due	Tier	Class	Earliest safe	Latest safe	Basis for the classification
Comenity	\$65.00	monthly	12 Sep	3	UNKNOWN	16 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
One Main loan	\$388.24	monthly	6 Sep	3	FIXED	10 Aug	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
PNC	\$150.00	monthly	8 Sep	3	UNKNOWN	12 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Upgrade basement	\$86.36	monthly	18 Aug	3	FIXED	22 Jul	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
Upstart debt consol	\$161.00	monthly	15 Sep	3	FIXED	19 Aug	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
Upstart past bills	\$86.46	monthly	24 Aug	3	FIXED	28 Jul	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
Upstart past bills (from Sep)	\$337.61	monthly	24 Sep	3	FIXED	28 Aug	not established	Recorded as non-deferrable in bills.json - missing it has a consequence the household has already flagged.
Loew & Patel braces	\$400.00	monthly	15 Sep	4	UNKNOWN	19 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
NJ Total Select 1of4	\$517.50	once	6 Oct	4	UNKNOWN	7 Oct	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
NJ Total Select 2of4	\$517.50	once	-	4	UNKNOWN	-	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
NJ Total Select 3of4	\$517.50	once	-	4	UNKNOWN	-	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
NJ Total Select 4of4	\$517.50	once	-	4	UNKNOWN	-	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.

Bill	Forecast	Freq	Next due	Tier	Class	Earliest safe	Latest safe	Basis for the classification
Republic trash Q1	\$177.30	once	-	4	UNKNOWN	-	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Republic trash Q4	\$177.30	once	7 Oct	4	UNKNOWN	8 Oct	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Culligan	\$307.70	monthly	4 Sep	5	UNKNOWN	8 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Home Depot	\$100.00	monthly	14 Sep	5	UNKNOWN	18 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Lowes	\$126.00	monthly	18 Aug	5	UNKNOWN	22 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
Netflix	\$21.31	monthly	23 Aug	5	UNKNOWN	27 Jul	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.
TJX Rewards	\$36.00	monthly	4 Sep	5	UNKNOWN	8 Aug	not established	Marked deferrable, but no due date, grace period or late-fee term is recorded. How many days it may safely slip is not derivable from the repository.

Amber rows are the confirmations that would unlock optimisation. 'Latest safe' reads 'not established' wherever the repository does not record one - that is a missing fact, not a judgement that the bill cannot move.

5. Current cash-flow schedule, day by day

Every day from Sun 16 Aug to Wed 14 Oct. Opening balance, money in, every obligation out, and the closing balance after each. Rows shaded red close below zero; amber rows close below \$750.00, the large_payment_threshold from config.json, meaning the day could not absorb one large obligation.

Date	Opening	In	Money in	Bills / expenses out	Out	Closing	Risk	Action
Sun 16 Aug	\$2,414.80	-	-	Groceries \$319.09; Fuel \$90.77	\$409.86	\$2,004.94	OK	
Mon 17 Aug	\$2,004.94	-	-	Capital One CC \$163.00; Chase CC Amazon \$136.00	\$299.00	\$1,705.94	OK	
Tue 18 Aug	\$1,705.94	-	-	CCU car loan \$306.90; Comcast \$138.93; Lowes \$126.00; Upgrade basement \$86.36	\$658.19	\$1,047.75	OK	
Wed 19 Aug	\$1,047.75	\$2,016.06	Insight Global payroll \$2,016.06	IRS state \$114.00; IRS federal \$103.00	\$217.00	\$2,846.81	OK	
Thu 20 Aug	\$2,846.81	-	-	-	-	\$2,846.81	OK	
Fri 21 Aug	\$2,846.81	-	-	-	-	\$2,846.81	OK	
Sat 22 Aug	\$2,846.81	-	-	Chase CC United \$88.00	\$88.00	\$2,758.81	OK	
Sun 23 Aug	\$2,758.81	-	-	Groceries \$319.09; Fuel \$90.77; Netflix \$21.31	\$431.17	\$2,327.64	OK	
Mon 24 Aug	\$2,327.64	-	-	Mortgage \$1,801.62; Kia \$742.65; Upstart past bills \$86.46	\$2,630.73	-\$303.09	NEGA...	Short \$303.09 - see section 8
Tue 25 Aug	-\$303.09	-	-	-	-	-\$303.09	NEGA...	
Wed 26 Aug	-\$303.09	\$2,016.06	Insight Global payroll \$2,016.06	-	-	\$1,712.97	OK	
Thu 27 Aug	\$1,712.97	-	-	Barclay Old Navy \$84.78	\$84.78	\$1,628.19	OK	
Fri 28 Aug	\$1,628.19	-	-	-	-	\$1,628.19	OK	
Sat 29 Aug	\$1,628.19	-	-	Xfinity Mobile \$88.82; Affirm \$73.07	\$161.89	\$1,466.30	OK	
Sun 30 Aug	\$1,466.30	-	-	Groceries \$319.09; Fuel \$90.77	\$409.86	\$1,056.44	OK	
Mon 31 Aug	\$1,056.44	-	-	-	-	\$1,056.44	OK	
Tue 01 Sep	\$1,056.44	-	-	Sewer \$304.50	\$304.50	\$751.94	OK	
Wed 02 Sep	\$751.94	\$2,016.06	Insight Global payroll \$2,016.06	-	-	\$2,768.00	OK	
Thu 03 Sep	\$2,768.00	-	-	JCP&L / FirstEnergy \$374.22	\$374.22	\$2,393.78	OK	
Fri 04 Sep	\$2,393.78	-	-	Culligan \$307.70; TJX Rewards \$36.00	\$343.70	\$2,050.08	OK	

Date	Opening	In	Money in	Bills / expenses out	Out	Closing	Risk	Action
Sat 05 Sep	\$2,050.08	-	-	-	-	\$2,050.08	OK	
Sun 06 Sep	\$2,050.08	-	-	One Main loan \$388.24; Groceries \$319.09; Fuel \$90.77	\$798.10	\$1,251.98	OK	
Mon 07 Sep	\$1,251.98	-	-	Mortgage \$1,801.62	\$1,801.62	-\$549.64	NEGA...	
Tue 08 Sep	-\$549.64	-	-	PNC \$150.00	\$150.00	-\$699.64	NEGA...	Short \$699.64 - see section 8
Wed 09 Sep	-\$699.64	\$2,016.06	Insight Global payroll \$2,016.06	-	-	\$1,316.42	OK	
Thu 10 Sep	\$1,316.42	-	-	-	-	\$1,316.42	OK	
Fri 11 Sep	\$1,316.42	-	-	Elizabethtown Gas \$60.01; T-Mobile Lia \$20.75	\$80.76	\$1,235.66	OK	
Sat 12 Sep	\$1,235.66	-	-	Hanover Auto \$513.27; Comenity \$65.00	\$578.27	\$657.39	TIGHT	
Sun 13 Sep	\$657.39	-	-	Groceries \$319.09; Fuel \$90.77	\$409.86	\$247.53	TIGHT	
Mon 14 Sep	\$247.53	-	-	Home Depot \$100.00	\$100.00	\$147.53	TIGHT	
Tue 15 Sep	\$147.53	-	-	Loew & Patel braces \$400.00; Upstart debt consol \$161.00	\$561.00	-\$413.47	NEGA...	Short \$413.47 - see section 8
Wed 16 Sep	-\$413.47	\$2,016.06	Insight Global payroll \$2,016.06	-	-	\$1,602.59	OK	
Thu 17 Sep	\$1,602.59	-	-	Capital One CC \$163.00; Chase CC Amazon \$136.00	\$299.00	\$1,303.59	OK	
Fri 18 Sep	\$1,303.59	-	-	CCU car loan \$306.90; Comcast \$138.93; Lowes \$126.00; Upgrade basement \$86.36	\$658.19	\$645.40	TIGHT	
Sat 19 Sep	\$645.40	-	-	Town of Clinton water \$873.53; IRS state \$114.00; IRS federal \$103.00	\$1,090.53	-\$445.13	NEGA...	
Sun 20 Sep	-\$445.13	-	-	Groceries \$319.09; Fuel \$90.77	\$409.86	-\$854.99	NEGA...	
Mon 21 Sep	-\$854.99	-	-	Mortgage \$1,801.62	\$1,801.62	-\$2,656.61	NEGA...	
Tue 22 Sep	-\$2,656.61	-	-	Chase CC United \$88.00	\$88.00	-\$2,744.61	NEGA...	
Wed 23 Sep	-\$2,744.61	\$2,016.06	Insight Global payroll \$2,016.06	Netflix \$21.31	\$21.31	-\$749.86	NEGA...	
Thu 24 Sep	-\$749.86	-	-	Kia \$742.65; Upstart past bills (from Sep) \$337.61	\$1,080.26	-\$1,830.12	NEGA...	
Fri 25 Sep	-\$1,830.12	-	-	-	-	-\$1,830.12	NEGA...	
Sat 26 Sep	-\$1,830.12	-	-	-	-	-\$1,830.12	NEGA...	

Date	Opening	In	Money in	Bills / expenses out	Out	Closing	Risk	Action
Sun 27 Sep	-\$1,830.12	-	-	Groceries \$319.09; Fuel \$90.77; Barclay Old Navy \$84.78	\$494.64	-\$2,324.76	NEGA...	
Mon 28 Sep	-\$2,324.76	-	-	-	-	-\$2,324.76	NEGA...	
Tue 29 Sep	-\$2,324.76	-	-	Xfinity Mobile \$88.82; Affirm \$73.07	\$161.89	-\$2,486.65	NEGA...	
Wed 30 Sep	-\$2,486.65	\$2,016.06	Insight Global payroll \$2,016.06	-	-	-\$470.59	NEGA...	
Thu 01 Oct	-\$470.59	-	-	-	-	-\$470.59	NEGA...	
Fri 02 Oct	-\$470.59	-	-	-	-	-\$470.59	NEGA...	
Sat 03 Oct	-\$470.59	-	-	JCP&L / FirstEnergy \$295.83	\$295.83	-\$766.42	NEGA...	
Sun 04 Oct	-\$766.42	-	-	Groceries \$319.09; Culligan \$307.70; Fuel \$90.77; TJX Rewards \$36.00	\$753.56	-\$1,519.98	NEGA...	
Mon 05 Oct	-\$1,519.98	-	-	Mortgage \$1,801.62	\$1,801.62	-\$3,321.60	NEGA...	
Tue 06 Oct	-\$3,321.60	-	-	NJ Total Select 1of4 \$517.50; One Main loan \$388.24	\$905.74	-\$4,227.34	NEGA...	Short \$4,227.34 - see section 8
Wed 07 Oct	-\$4,227.34	\$2,016.06	Insight Global payroll \$2,016.06	Republic trash Q4 \$177.30	\$177.30	-\$2,388.58	NEGA...	
Thu 08 Oct	-\$2,388.58	-	-	PNC \$150.00	\$150.00	-\$2,538.58	NEGA...	
Fri 09 Oct	-\$2,538.58	-	-	-	-	-\$2,538.58	NEGA...	
Sat 10 Oct	-\$2,538.58	-	-	-	-	-\$2,538.58	NEGA...	
Sun 11 Oct	-\$2,538.58	-	-	Groceries \$319.09; Fuel \$90.77; Elizabethtown Gas \$69.16; T-Mobile Lia \$20.75	\$499.77	-\$3,038.35	NEGA...	
Mon 12 Oct	-\$3,038.35	-	-	Hanover Auto \$513.27; Comenity \$65.00	\$578.27	-\$3,616.62	NEGA...	
Tue 13 Oct	-\$3,616.62	-	-	-	-	-\$3,616.62	NEGA...	
Wed 14 Oct	-\$3,616.62	\$2,016.06	Insight Global payroll \$2,016.06	Home Depot \$100.00	\$100.00	-\$1,700.56	NEGA...	

Deferred occurrences are excluded from the outflow columns because they do not leave the account on that date; they remain owed and are listed in section 9.

6. Optimised schedule

There is no optimised schedule to show, because the repository does not support a single legitimate date change. Every obligation is either FIXED or UNKNOWN, and this report will not move a bill later on the basis that it technically could be paid later. What follows instead is the schedule that becomes available the moment those windows are confirmed.

Conditional schedule - assumes every UNKNOWN bill may slip up to 7 days

Conditional, and not a recommendation

The plan below exists to size the prize, not to be acted on. It assumes a uniform tolerance that nothing in the repository establishes. Confirm the individual windows in section 4 and this becomes a real schedule; until then it is arithmetic on an assumption, clearly labelled as one.

Bill	Amount	From	To	Days	Tier	Confidence
Chase CC United	\$88.00	22 Aug	26 Aug	+4	3	LOW
Netflix	\$21.31	23 Aug	26 Aug	+3	5	LOW
Culligan	\$307.70	4 Sep	9 Sep	+5	5	LOW
TJX Rewards	\$36.00	4 Sep	9 Sep	+5	5	LOW
PNC	\$150.00	8 Sep	9 Sep	+1	3	LOW
Home Depot	\$100.00	14 Sep	16 Sep	+2	5	LOW
Loew & Patel braces	\$400.00	15 Sep	16 Sep	+1	4	LOW

Measure	Current	Conditional	Change
Lowest projected balance	-\$4,227.34	-\$4,227.34	\$0.00
Days closing below zero	31	30	-1
Bills moved	-	7	7
Dollars rescheduled	-	\$1,103.01	\$1,103.01

Even on that assumption the lowest projected balance moves from -\$4,227.34 to -\$4,227.34 and 30 days still close below zero. Rescheduling is not the answer to this projection; it is at best a partial mitigation of the earliest episodes.

7. Recommended bill-date changes

None. No bill in this repository has a recorded latest-safe payment date, so there is no change this report can recommend on evidence. Recommending one anyway would mean inventing the single fact that decides whether it is safe - the deadline. The actionable request is therefore the confirmation itself, listed below in the order the projection would benefit.

Bill	Amount	Currently paid	Would need	Question to answer
Chase CC United	\$88.00	22 Aug	+4 days	What is the actual due date, and what happens if it is paid after it?
Netflix	\$21.31	23 Aug	+3 days	What is the actual due date, and what happens if it is paid after it?
Culligan	\$307.70	4 Sep	+5 days	What is the actual due date, and what happens if it is paid after it?
TJX Rewards	\$36.00	4 Sep	+5 days	What is the actual due date, and what happens if it is paid after it?
PNC	\$150.00	8 Sep	+1 days	What is the actual due date, and what happens if it is paid after it?
Home Depot	\$100.00	14 Sep	+2 days	What is the actual due date, and what happens if it is paid after it?
Loew & Patel braces	\$400.00	15 Sep	+1 days	What is the actual due date, and what happens if it is paid after it?

Answering these turns section 6's conditional schedule into a set of real recommendations. Record the answer in bills.json as a payment_window entry and the engine will use it automatically.

8. Shortfalls that no date change can fix

Each of these is a period where the account closes below zero and the obligations landing in it cannot legitimately be moved out of it. Moving dates redistributes a deficit; it does not fund one.

Mon 24 Aug to Tue 25 Aug - short \$303.09

Field	Value
Worst day	Mon 24 Aug
Projected balance	-\$303.09
Cash needed to hold at zero	\$303.09
Next income after it	Wed 26 Aug for \$2,016.06
Largest obligations in the period	Mortgage \$1,801.62 on 24 Aug; Kia \$742.65 on 24 Aug; Upstart past bills \$86.46 on 24 Aug
Why rescheduling does not solve it	Nothing due on or before 25 Aug has a payment date that may legitimately move later, so no schedule change closes this gap.

Options that do address it, in rough order of how little they cost: confirm whether any obligation in the period may legitimately be paid later; reduce discretionary spending in the days before it by the amount shown; bring forward or restore income; or arrange the cash from outside. This report does not recommend borrowing to close a modelled gap, because the gap is partly a modelling question and section 10 shows by how much.

Mon 7 Sep to Tue 8 Sep - short \$699.64

Field	Value
Worst day	Tue 8 Sep
Projected balance	-\$699.64
Cash needed to hold at zero	\$699.64
Next income after it	Wed 9 Sep for \$2,016.06
Largest obligations in the period	Mortgage \$1,801.62 on 7 Sep; PNC \$150.00 on 8 Sep
Why rescheduling does not solve it	Nothing due on or before 8 Sep has a payment date that may legitimately move later, so no schedule change closes this gap.

Options that do address it, in rough order of how little they cost: confirm whether any obligation in the period may legitimately be paid later; reduce discretionary spending in the days before it by the amount shown; bring forward or restore income; or arrange the cash from outside. This report does not recommend borrowing to close a modelled gap, because the gap is partly a modelling question and section 10 shows by how much.

Tue 15 Sep to Tue 15 Sep - short \$413.47

Field	Value
Worst day	Tue 15 Sep
Projected balance	-\$413.47
Cash needed to hold at zero	\$413.47
Next income after it	Wed 16 Sep for \$2,016.06
Largest obligations in the period	Loew & Patel braces \$400.00 on 15 Sep; Upstart debt consol \$161.00 on 15 Sep
Why rescheduling does not solve it	Nothing due on or before 15 Sep has a payment date that may legitimately move later, so no schedule change closes this gap.

Options that do address it, in rough order of how little they cost: confirm whether any obligation in the period may legitimately be paid later; reduce discretionary spending in the days before it by the amount shown; bring forward or restore income; or arrange the cash from outside. This report does not recommend borrowing to close a modelled gap, because the gap is partly a modelling question and section 10 shows by how much.

Sat 19 Sep to Wed 14 Oct - short \$4,227.34

Field	Value
Worst day	Tue 6 Oct
Projected balance	-\$4,227.34
Cash needed to hold at zero	\$4,227.34
Next income after it	Wed 7 Oct for \$2,016.06
Largest obligations in the period	Mortgage \$1,801.62 on 21 Sep; Mortgage \$1,801.62 on 5 Oct; Town of Clinton water \$873.53 on 19 Sep; Kia \$742.65 on 24 Sep; NJ Total Select 1of4 \$517.50 on 6 Oct; Hanover Auto \$513.27 on 12 Oct
Why rescheduling does not solve it	Nothing due on or before 14 Oct has a payment date that may legitimately move later, so no schedule change closes this gap.

Options that do address it, in rough order of how little they cost: confirm whether any obligation in the period may legitimately be paid later; reduce discretionary spending in the days before it by the amount shown; bring forward or restore income; or arrange the cash from outside. This report does not recommend borrowing to close a modelled gap, because the gap is partly a modelling question and section 10 shows by how much.

9. Committed obligations versus discretionary spending

Everything above counts committed obligations only. No everyday spending is assumed anywhere in this report, which is deliberate: an invented spending rate charged to the household is an assumption wearing the costume of arithmetic. What the projection can carry is derived from it instead.

Headroom over 30 days	Binding day	Sustainable per day
-\$699.64	8 Sep	-\$33.68

There is no discretionary capacity at all

Headroom is -\$699.64, binding on Tue 8 Sep. That figure carries bills and income only. It is already negative before a single dollar of groceries beyond the modelled weekly amount, fuel, or anything else - so every additional dollar spent deepens the worst day by exactly that dollar. Note that groceries and fuel ARE modelled as committed weekly obligations here, at their historical means, so they are not double counted in this figure.

10. Data integrity findings

Differences between what the repository stores and what the engine forecasts. Neither is automatically wrong - forecasting a variable bill at its worst observed value is a deliberate rule - but each one is a place where the displayed model and the stored model disagree, and each moves the projection.

Bill	Stored versus forecast	Classification
Capital One CC	stored \$105.00, forecast \$163.00 (\$58.00 per occurrence)	SUPERSEDED
Chase CC Amazon	stored \$80.00, forecast \$136.00 (\$56.00 per occurrence)	SUPERSEDED
Lowes	stored \$69.00, forecast \$126.00 (\$57.00 per occurrence)	SUPERSEDED
Chase CC United	stored \$46.00, forecast \$88.00 (\$42.00 per occurrence)	SUPERSEDED
PNC	stored \$70.00, forecast \$150.00 (\$80.00 per occurrence)	FORECAST-AT-WORST
Barclay Old Navy	stored \$60.74, forecast \$84.78 (\$24.04 per occurrence)	FORECAST-AT-WORST
Comenity	stored \$50.00, forecast \$65.00 (\$15.00 per occurrence)	FORECAST-AT-WORST
Culligan	stored \$40.39, forecast \$307.70 (\$267.31 per occurrence)	FORECAST-AT-WORST
Town of Clinton water	stored \$294.03, forecast \$873.53 (\$579.50 per occurrence)	SUPERSEDED

SUPERSEDED means the household set the current amount and the engine is forecasting a higher pre-correction observation over the top of it. FORECAST-AT-WORST is the documented rule applying as intended - shown so its size is visible.

Sensitivity - what each modelling decision is worth

Scenario	What it changes	Lowest balance	Days below ...
As configured today	Every setting exactly as the live engine runs it.	-\$4,227.34	31
Honour 5 household-corrected amount(s)	Stop forecasting renegotiated bills at their pre-renegotiation observed maximum.	-\$3,221.84	24
All variable bills at their stated amount	Removes the forecast-at-worst rule entirely. The optimistic bound, shown only to bracket the answer.	-\$2,544.14	21
Reactivate: Branchburg BOE payroll	Every income source currently switched off is counted again, at its recorded amount and cadence.	\$1,047.75	0
Both corrections together	Household-corrected amounts honoured AND the inactive income restored.	\$1,218.75	0

Each row re-runs the whole projection with one decision changed and everything else held constant. These are diagnostics, not proposals - none of them has been applied to the live model.

11. Assumptions and unknowns

Labelled honestly. Anything the repository does not establish is marked as needing confirmation rather than filled in.

Payment flexibility is not recorded anywhere

24 active obligations are marked deferrable, but nothing in bills.json records a due date separate from the payment date, a grace period, or a late-fee term. Until those are supplied, no bill can be moved later on evidence. This is the single largest gap between what this report can prove and what it could recommend.

Income unverified: Branchburg BOE payroll

Flagged needs_review in income.json. Edited in the app 13 Aug 2026: deactivated. Previous note: Amount corrected from \$2,840.34. Observed \$2,829.31 on 04/13, 04/28, 05/13, 05/27 and 06/10/2026, exactly 14-15 days apart. A \$2,837.55 credit on 06/16 breaks the pattern and may be

Income switched off: Branchburg BOE payroll

\$2,829.31 biweekly is recorded but not counted. Edited in the app 13 Aug 2026: deactivated. Previous note: Amount corrected from \$2,840.34. Observed \$2,829.31 on 04/13, 04/28, 05/13, 05/27 and 06/10/2026, exactly 14-15 days apart. A \$2,837.55 credit on 06/16 breaks the pattern and may be

Income unverified: Insight Global payroll

Flagged needs_review in income.json. CORRECTED. Was \$2,853.17/week; the actual employer is Insight Global and the modal weekly deposit is \$2,016.06. That is \$837.11/week overstated, roughly \$3,627/month of income the projection was counting and never receiving. Deposits range

Income unverified: Charlie and Clark

Flagged needs_review in income.json. Added in the app 13 Aug 2026: once, first on 13 Aug 2026. Not yet seen on a statement.

Bill needs review: Mortgage

CORRECTED from semimonthly to biweekly. 15 consecutive payments 12/15/2025-06/29/2026 are exactly 14 days apart. Semimonthly modelled 24/yr; biweekly is 26/yr, a \$3,603.24/yr difference. Extra same-week entries in the statements are failed-payment retries, not

Bill needs review: Kia

Amount corrected from \$578.64. Actually recurs every 28 days (05/04, 06/01, 06/29), not monthly, so it drifts earlier and yields 13 payments in some years. due_day 24 tracks the next two occurrences; revisit each quarter.

Bill needs review: PNC

Varies \$70.00-\$150.00 and the date moves. \$70.00 is the recent floor.

Bill needs review: Comenity

Irregular: 12/16, 02/03, 03/13, 05/05, 06/12, amounts \$40.00-\$65.00. Not on the household sheet.

Bill needs review: Culligan

Base \$40.39, but 01/05 was \$307.70 and 04/06 was \$143.39 - likely periodic salt or service deliveries on top.

Bill needs review: Republic trash Q4

CORRECTED from biweekly \$92.00 to QUARTERLY \$177.30. Observed 01/07 \$168.12 and 04/07 \$177.30, exactly 90 days apart. Modelled as dated one-offs because the schema has no quarterly frequency.

Bill needs review: Republic trash Q1

See trash. Next quarterly instalment.

Bill needs review: Sewer

POSSIBLY ANNUAL, UNCONFIRMED - the household does not yet know. No matching payment appears in 19 months of statements, including the September window. Left active at \$304.50 on 1 Sep so it is budgeted for rather than forgotten.

Bill needs review: Upstart past bills (from Sep)

Steps up from \$86.46 on 24 Sep 2026 per the household. The household said the reduced rate ran 'two months'; if that means it

ends after 24 Aug rather than 24 Sep, this start date is a month late and the forecast is optimistic by \$251.15 for September.
Confirm

Bill needs review: Town of Clinton water

Actual annual spend: 2022 \$1,088.42, 2023 \$1,121.76, 2024 \$1,318.11 (mean \$1,176.10). Billed irregularly - between two and five payments a year, not on a fixed date - so this is the annual mean spread over four quarters rather than a real due date.

Bill needs review: IRS state

New, \$114.00/month, per the household. Due day assumed the same as the federal instalment; confirm. Not yet visible in any statement.

Bill needs review: Loew & Patel braces

SCHEDULE UNKNOWN - flagged, not guessed. The statements show \$1,000.00 on 06/10/2026, \$129.47 on 04/29 and 06/24, and smaller amounts through the spring. No pattern is derivable. Held at the household's \$400.00/month placeholder.
observed_max is deliberately N

12. Methodology

The projection starts from the last recorded balance and walks forward one day at a time. For each day: opening balance, plus income landing that day, minus obligations landing that day, equals closing balance, which becomes the next day's opening. Income is ordered before outflows within a day, because assuming the reverse would flag a phantom shortfall on every payday. No allowance for everyday spending is charged. All money is handled as exact decimal, rounded half up; floating point is never used, because a projection is a long chain of additions.

Occurrence dates come from one calendar module used by every other part of the system, so the report and the app cannot drift apart. Monthly bills clamp to the last day of a short month. Weekly and biweekly items step 7 or 14 days from an anchor date, which is what makes a biweekly mortgage produce 26 payments a year rather than 24. Nothing is shifted off a weekend: a due date landing on a Saturday is counted on the Saturday, because shifting it to Monday would make the projection optimistic on precisely the days that matter.

Optimisation works on episodes. The first stretch of days closing below the floor is identified, and its deficit is the amount that must be kept out of it. Candidates are obligations landing on or before the end of that stretch whose recorded window permits landing after it. They are taken least-critical first and largest first, so the fewest moves close the gap, and any move the episode turns out not to need is then dropped. The curve is re-run and the next episode examined, until none remain or nothing legitimate is left to move.

Moving a payment later is safe for the balance curve in a precise sense: shifting an obligation from day d to day t raises every day from d to $t-1$ by its amount and changes no day from t onward, because the money has left by then either way. So a later date never worsens any day. The entire cost of moving is the risk of paying late, which is why a recorded deadline is required before this report will propose it.

Total outflow is identical before and after optimisation, and that is asserted in the test suite rather than assumed. A schedule that reduces what is owed is not a schedule.